

ENGINEERING WORK SAMPLE

HELPLESK MANAGEMENT SYSTEM

A Full-Stack Developer Productivity Notepad, Task Tracking System, and Client Update Dispatcher built with CakePHP 5 and MySQL.

PROJECT TYPE

Full-Stack Web App

PRIMARY STACK

PHP 8.2 • CakePHP 5.4

DATABASE

MySQL 8 (InnoDB)

SOURCE CODE

[GitHub Repo](#)

Project Overview & Description

Application Purpose & Problem Solved

[Verified Source Code](#)

The Helpdesk Management System is a specialized web application engineered for software developers and engineering teams to track daily tasks, organize work notes by project and client, and generate client-ready daily status reports. It solves the everyday challenge of fragmented developer work logging and time-consuming manual email drafting by unifying task entry, automated task-count calculations, and real-time email previews into a single responsive workspace. The system incorporates an integrated Google Gemini AI text polishing proxy to refine grammar, standardize bulleted layouts, and translate Gujarati or Gujlish notes into professional English while strictly preserving structural indentation. Built specifically for freelance engineers, agency developers, and technical leads managing multi-client deliverables, the platform streamlines the development journal lifecycle from initial draft to verified database record and dispatched SMTP email.

Centralized Developer Journal

Allows developers to maintain a chronological diary of coding tasks with automatic categorization by client, project, and work date.

Client Communication Automation

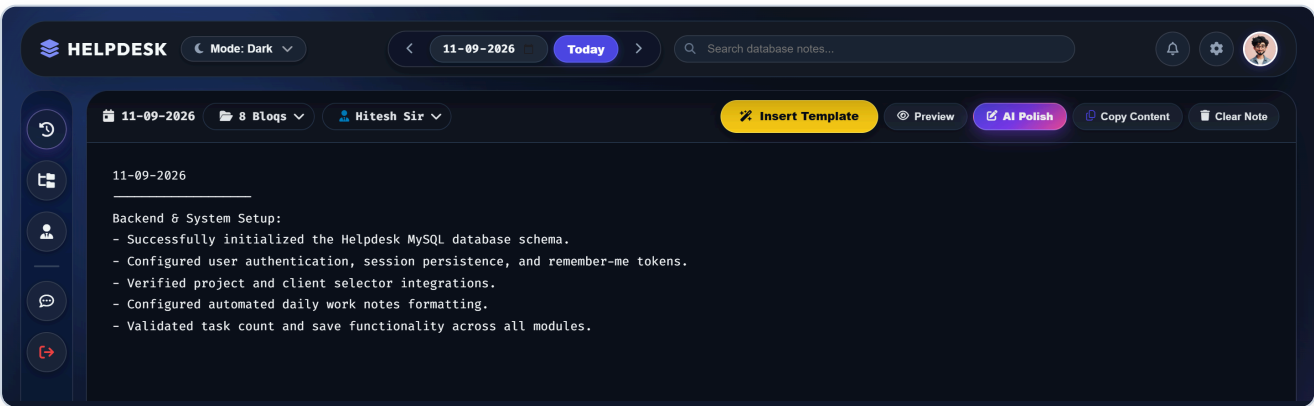
Converts raw daily task points into structured client progress reports with automated recipient routing and live HTML email generation.

AI-Assisted Text Refinement

Leverages a Google Gemini proxy to correct technical grammar, standardize vocabulary, and translate regional notes without altering formatting.

Multi-Tenant Client Management

Features full CRUD management for client entities and project workspaces with user-level isolation and default selector preferences.



Technologies Used

Backend & Core Framework

MVC

PHP 8.1+ / 8.2: Strict types, modern object-oriented syntax.
CakePHP 5.4.1: Modern rapid application framework utilizing MVC architecture, ORM, dependency injection, and middleware.
CakePHP Mailer: Native SmtplibTransport delivery engine.
Composer: Dependency manager and PSR-4 autoloader.

Database & Data Persistence

Relational

MySQL 8.x: Relational storage engine with InnoDB tables.
utf8mb4 Collation: Multilingual Unicode character support.
Foreign Key Constraints: ON DELETE CASCADE & SET NULL.
Composite Indexing: Unique constraints on date/project/user.

Frontend & UI Styling

Custom CSS

Vanilla CSS3: Custom Glassmorphism design tokens.
Dual-Theme System: Instant Light and Dark mode toggle.
Layouts: Responsive Flexbox and multi-column CSS Grid.
Typography: Google Fonts (Inter, Outfit, JetBrains Mono).

Security & Cryptography

OpenSSL

AES-256-CBC: Encrypted storage for user API keys & SMTP passwords.
bcrypt (PASSWORD_DEFAULT): Salted one-way password hashing.
HMAC SHA-256: Cryptographic 30-day remember-me tokens.
Session Isolation: Protected controller beforeFilter gatekeeping.

External APIs & Integrations

AI & Auth

Google Gemini API: Generative Language API v1beta proxy.
Google OAuth 2.0: Google Identity Services (GIS) auth flow.
Multi-Provider SMTP: Gmail, Hostinger, and cPanel webmail.
FontAwesome 6: Consistent vector interface iconography.

Testing & Code Quality

Quality

PHPUnit 10: Unit and integration test orchestration.
PHPStan: Static code analysis and type safety verification.
CakePHP CodeSniffer: PSR-12 coding standard enforcement.
Git: Feature branching and version control history.

Project Structure & Architecture

Multi-Tier Architectural Dataflow

CakePHP 5 Pipeline

PRESENTATION	Server-rendered CakePHP View templates (Tasks/index.php, DailyUpdates/index.php, Users/login.php) + Vanilla JavaScript (Fetch API, DOM manipulation).
MIDDLEWARE	RoutingMiddleware, BodyParserMiddleware, ErrorHandlerMiddleware, and ApplicationController authentication filters with remember-me validation.
CONTROLLERS	TasksController (notepad & AI proxy), DailyUpdatesController (report compiler & email), UsersController (auth & profile), ClientsController, ProjectsController.
SERVICES & ORM	EmailService (SMTP dispatch & validation), Table classes (ORM relations & queries), Entity classes (data mutators, AES-256 encryption, password hashing).
DATA & APIS	MySQL 8.x Database Engine • Google Gemini AI REST API (Generative Language v1beta) • Google OAuth 2.0 GIS • Gmail / Hostinger SMTP Servers.

src/Controller/

Coordinates HTTP/AJAX requests, handles request payload validation, enforces session/cookie auth, and dispatches JSON responses.

src/Model/

Table classes manage database schemas, associations (hasMany, belongsTo), and entities encapsulate encryption and hashing mutators.

src/Service/

Decoupled EmailService encapsulates multi-transport SMTP routing, recipient parsing, HTML template layout, and OTP dispatch.

Key Features Verified from Source Code

FEATURE 01

Daily Work Log Notepad

A centralized real-time editor designed for drafting and persisting daily development tasks. Supports automated formatting, date navigation, and instant persistence linked to distinct projects and clients.

TasksController.php • WorkLogsTable

FEATURE 02

AI-Powered Text Polish & Translation

Integrated Google Gemini proxy that corrects grammar, spelling, and punctuation while translating Gujarati or Gujlish notes into clean professional English with zero alteration to bullet indentation or layout.

Google Gemini API • TasksController::aiPolish

FEATURE 03

Daily Update & Email Generator

Automatically compiles completed tasks from the daily notepad into structured client updates categorized by done, in-progress, and remaining tasks, featuring real-time synchronized HTML preview.

DailyUpdatesController.php • Live Preview

FEATURE 04

Multi-Provider SMTP Dispatcher

Built-in email composer supporting Gmail, Hostinger, and custom SMTP transports with automatic To, Cc, and Bcc population based on saved client contacts and deliverability validation.

EmailService.php • CakePHP Mailer

FEATURE 05

Multi-Project & Client Management

Comprehensive CRUD system enabling developers to create, edit, and organize work logs across distinct client relationships and project workspaces with default selector preferences.

ProjectsController.php • ClientsController.php

FEATURE 06

Calendar & Work History Drawer

Collapsible side drawer with month navigation (Sep 2026), live numeric search filtering, and task count badges per historical entry, allowing instant inspection and editing of past logs.

TasksController::getMonthLogs • DOM Drawer

FEATURE 07

Global Database Search & Quick Filtering

Live search endpoint allowing developers to query past work notes and task entries across dates, project names, and keywords with instant highlight rendering in the user interface.

TasksController::globalSearch • MySQL LIKE

FEATURE 08

Authentication & Credential Encryption

Multi-layered security suite featuring bcrypt password hashing, AES-256-CBC encrypted credential storage for user API keys, 30-day remember-me cookie tokens, and Google OAuth 2.0 sign-in.

User.php (AES-256) • Google GIS Auth

Feature Implementation & Architecture Mapping

Model-View-Controller

Feature Module	Controller Action	Model / Service Layer	Database Table
Work Log Notepad	TasksController::index / saveLog	WorkLogsTable • WorkLog Entity	work_logs
AI Polish & Translation	TasksController::aiPolish	User::decryptString • cURL Gemini	users.api_key
Daily Client Updates	DailyUpdatesController::index	DailyUpdatesTable • DailyUpdate	daily_updates
SMTP Email Dispatch	DailyUpdatesController::sendEmail	App\Service\EmailService	EmailTransport
Projects & Clients CRUD	ProjectsController • ClientsController	ProjectsTable • ClientsTable	projects • clients
Security & Remember Me	UsersController::login / signup	User::_setPassword • HMAC SHA-256	users

Application Screenshots — Real Running Application

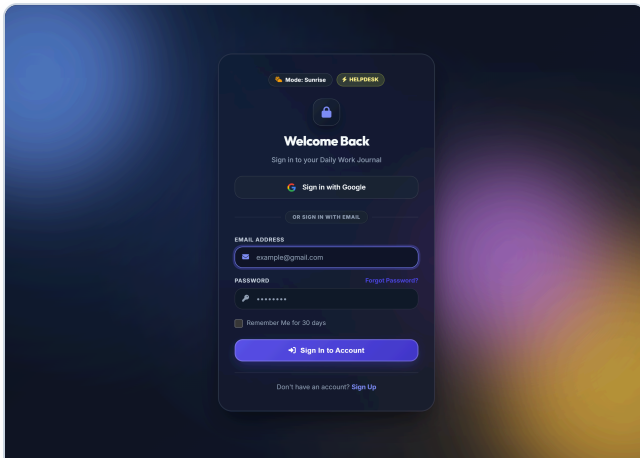


Figure 4.1: Sign-In & Authentication

Auth System

Glassmorphism sign-in view featuring Google OAuth 2.0, bcrypt authentication, remember-me persistent tokens, and dark/light theme toggle.

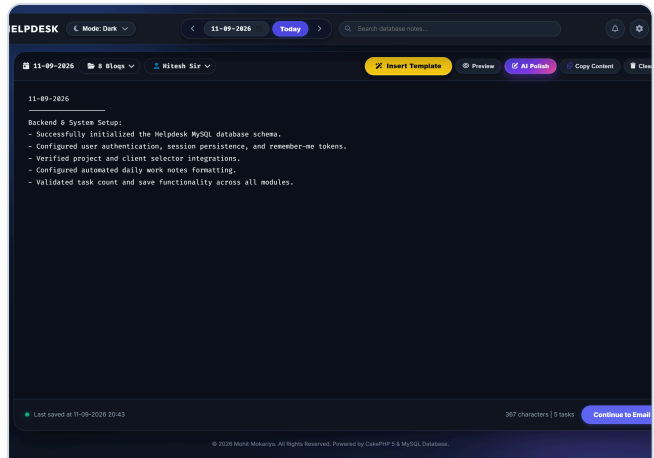


Figure 4.2: Daily Work Notepad

Core Notepad

Active developer workspace for 11-09-2026 showing project selector ('8 Bloqs'), client selector ('Hitesh Sir'), task editor, and 'AI Polish' action.

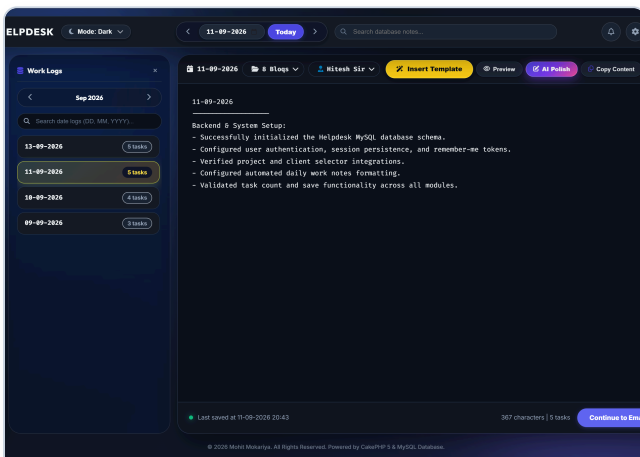


Figure 4.3: Work Logs Drawer

Log History

Animated slide-out drawer displaying month-wise log entries for September 2026, live numeric search filter, and task count badges per historical entry.

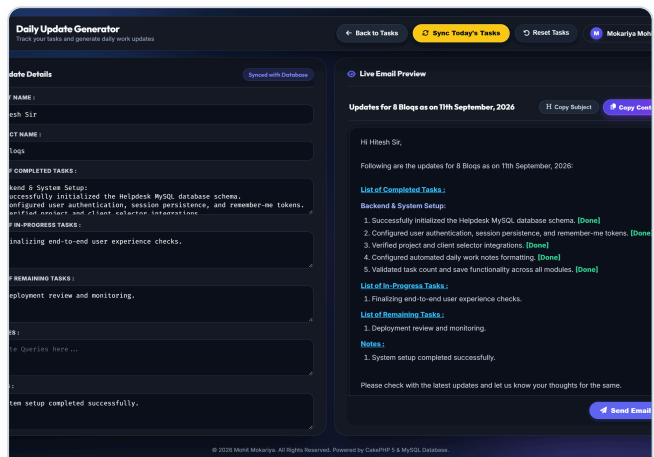


Figure 4.4: Daily Update Generator

Report Compiler

Update builder with categorized tasks synced from MySQL on the left, paired with an instantaneous real-time HTML email preview on the right.

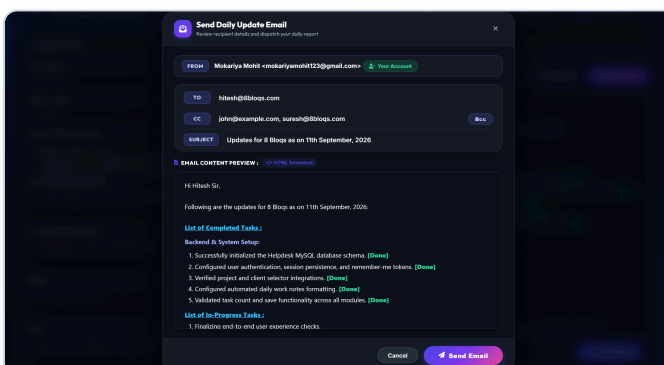


Figure 4.5: SMTP Email Composition & Dispatch Dialog

SMTP Email

Modal dispatch dialog demonstrating automated recipient routing (To: hitesh@8bloqs.com, Cc: team contacts), authenticated developer sender identity, subject line compilation, and formatted HTML body ready for immediate multi-provider SMTP delivery.

Development & Technical Highlights

1. Relational Schema Architecture & Data Integrity

[MySQL InnoDB](#)

The persistence tier is engineered in MySQL using InnoDB tables with strict relational integrity constraints. Foreign keys enforce `ON DELETE CASCADE` for user-owned records and `ON DELETE SET NULL` for project/client associations to prevent orphan records. Composite unique keys—specifically `unique_date_project_user` on `work_logs` (`log_date`, `project_id`, `user_id`) and `unique_update_date_project_user` on `daily_updates`—guarantee idempotency and eliminate duplicate log entries per developer workspace.

2. End-to-End Cryptography & Credential Isolation

[OpenSSL AES-256-CBC](#)

Security is enforced at the entity level: external third-party API keys (e.g. Google Gemini) and private user SMTP credentials are encrypted before insertion into MySQL using `OpenSSL AES-256-CBC` with randomly generated 16-byte initialization vectors (IVs) and salted cryptographic keys. User passwords are automatically hashed with one-way salted `bcrypt` (`PASSWORD_DEFAULT`), and persistent logins utilize a SHA-256 HMAC cookie token to prevent credential exposure.

3. Google Gemini AI REST Integration & Proxy Pipeline

[AI Text Engineering](#)

A dedicated backend proxy (`TasksController::aiPolish`) mediates between the user client and Google's Generative Language API v1beta. The proxy performs automated model discovery via REST to verify available generation models on the user's Google AI Studio account, falling back across candidate models (`gemini-1.5-flash` , `gemini-2.0-flash` , and `gemini-pro`). Custom regex sanitization strips markdown fences, self-checking thoughts, and commentary while strictly preserving line-by-line bullet indentation.

4. Intelligent Task Parsing & Counter Algorithm

[Regex Parsing](#)

The system implements an automated task counting engine in `TasksController::calculateTaskCount` . The parser analyzes multi-line text input, dynamically discarding date headers (`DD-MM-YYYY`), horizontal dividers (`---` or `===`), and category markers (e.g. `Backend:` , `Frontend:`). It inspects indentation levels to prevent nested sub-points from inflating the top-level task metric, accurately computing net completed items.

5. Decoupled Multi-Transport SMTP Email Architecture

[EmailService.php](#)

Email dispatching is encapsulated in `src/Service/EmailService.php` , isolating delivery logic from HTTP controllers. The service supports Gmail SSL SMTP (port 465), Hostinger SMTP, cPanel Webmail, and native PHP mail fallback. It features RFC-compliant multi-address parsing for comma- and semicolon-separated strings, regex syntax verification for To/Cc/Bcc lists, and dynamic HTML email styling with sender identity binding.

6. Pure CSS Glassmorphism & Theme Switching Engine

[Frontend Architecture](#)

The user interface avoids heavy utility-class frameworks in favor of a bespoke Vanilla CSS design system built on CSS custom properties. It incorporates backdrop-blur filters (Glassmorphism), animated radial glow orbs, and seamless dual-theme toggling (Dark Mode and Sunrise Light Mode). Interactive components including side drawers, modal dialogs, and popovers are driven by lightweight Vanilla ES6+ and jQuery event listeners.

Project Repository & Codebase Access

Official GitHub Repository

<https://github.com/mokariya-mohit/Helpdesk>

OPEN SOURCE CODEBASE

Verified Repository Structure

src/Controller/ AppController, TasksController, DailyUpdatesController, UsersController, ClientsController, ProjectsController.	src/Model/Table/ WorkLogsTable, DailyUpdatesTable, UsersTable, ProjectsTable, ClientsTable with ORM relations and validation.	src/Model/Entity/ User.php with AES-256 encrypt/decrypt methods and bcrypt mutator, WorkLog.php, DailyUpdate.php.
src/Service/ EmailService.php implementing multi-provider SMTP, email syntax validation, and HTML update templates.	templates/ Domain views for Tasks (Notepad), DailyUpdates (Email Generator), Users (Login/Signup), and layout templates.	database_schema.sql Complete MySQL schema dump with table definitions, foreign keys, indexes, and seeded records.

Professional Verification & Closing Statement

Engineering Competency Statement

Proposal Work Sample

"This project demonstrates practical software development experience including application development, debugging, database interaction, backend/frontend integration, and implementation of business functionality. It highlights hands-on proficiency in CakePHP 5 MVC architecture, relational database modeling with MySQL, secure session and cookie authentication, AES-256-CBC credential encryption, backend-to-frontend integration via RESTful AJAX endpoints, Google Gemini Generative AI integration, multi-provider SMTP email dispatching, and high-performance vanilla frontend engineering."

Software Engineer Profile

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Quality & Authenticity Notice

Every technology, architectural layer, database table, and feature listed in this portfolio document is directly verifiable in the public repository source code. Screenshots were captured directly from the live, fully functional local application environment.